

For data & ML platform teams · not another model demo

THE JOB NOBODY DEMOS

Put the vision model in your pipeline, not your product.

A 450M-parameter VLM is too small to be the star of a feature. That is exactly why it is the right tool for the unglamorous job: understanding your **entire image corpus** — captioning, tagging, detecting, counting, triaging — on hardware you already own, over data that never leaves your network.

Every figure on this page is sourced to a primary document. Where this model is the *wrong* choice, this page says so. You are an engineer; you will check.

You already priced the three options. None of them fit the corpus.

Somewhere in your org sit 5–500 million images — claims photos, product shots, inspection frames, scanned media, app screenshots. Most are unlabeled, under-tagged, or described by a schema you have already outgrown. The shortlist:

Frontier API

Gemini Flash, GPT-4o-mini, Claude Haiku. Trivial to call — but you meter every image, you cannot send regulated data off-prem, and the per-image token bill is wildly uneven between vendors (more below).

The killer: **re-running the corpus means paying for all of it again.**

Self-host 3–7B

Qwen2.5-VL-3B/7B. Genuinely capable, and Apache-licensed at 7B. But it is heavy per image; your batch throughput is GPU-bound and finance feels every node. The 3B is also **Qwen-Research-licensed**, not Apache — read the fine print.

Silicon on-device

A model tuned for a phone feature on a partner's NPU. Built to make one handset interaction feel instant — not to grind a data lake at batch. Frozen to the silicon roadmap, not yours.

THE REFRAME

A product reads each image once. A pipeline re-reads the whole corpus — forever.

Every time the model improves, the taxonomy shifts, or a new downstream consumer shows up, a data pipeline re-processes the back catalog. That re-processing is where per-image pricing quietly becomes the largest line item in your ML budget — and where a model with $\approx \$0$ **marginal cost** stops being a discount and starts changing the architecture.

Once "look at this image" is free and local, you stop sampling and start processing *everything*, re-running on every schema change, enriching aggressively, and pointing it at data the API was never allowed to see.

You do not need this model to be brilliant. You need it:

- **Cheap enough** to run over everything, repeatedly

- **Good enough** to triage, tag, detect, and route
- **Small enough** to fine-tune to *your* taxonomy
- **Private enough** to point at data the API can't touch

At 450M parameters, LFM2.5-VL is all four. The interactive below is the whole argument in one number.

THE CORPUS MATH · INTERACTIVE

What does it cost to understand your whole corpus — and to do it again?

Set your corpus, your monthly inflow, and how often you re-process the whole archive. Costs are **input-tokens only** at published 2026 list prices; output tokens add more on the API side and nothing on-device. Drag anything.

Insurance claims archive

Retail / e-commerce imagery

Manufacturing inspection

Media / DAM library

Existing corpus

30M images



New images / month

3M



Full-corpus re-processes / year

3×



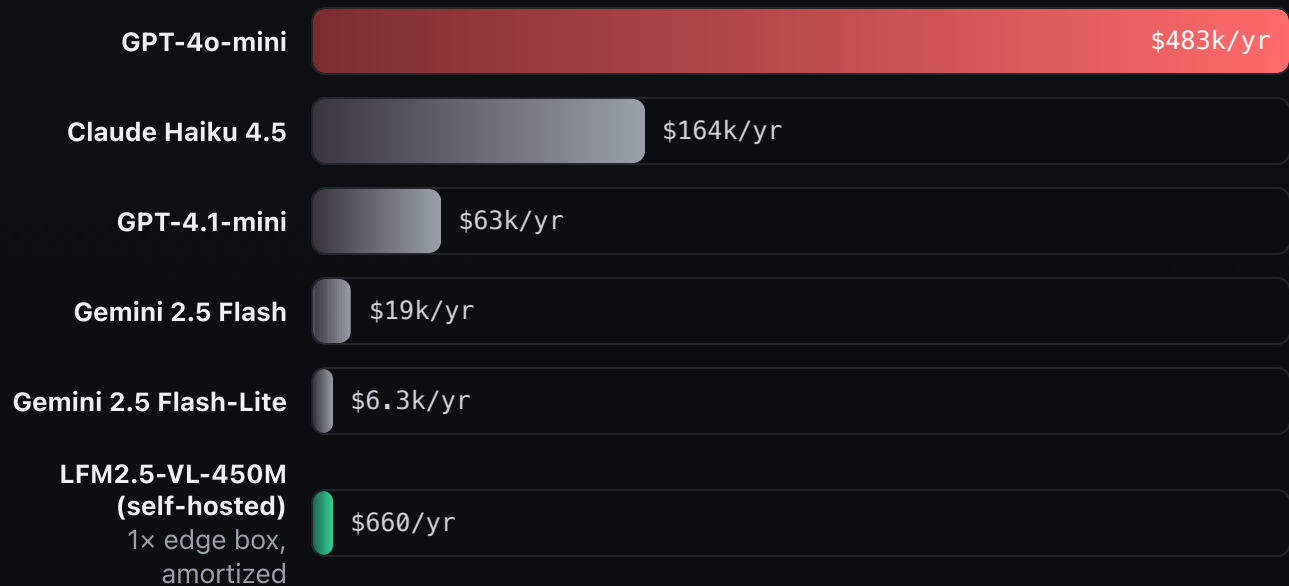
New taxonomy, model upgrade, new consumer, backfill. This is the lever product pitches ignore.

API price tier

GPT-4o-mini (\$3.83 / 1k img)



GPT-4o-mini's low sticker price hides a ~33× image-token multiplier — it is the most expensive option here.



Chosen API · per year

\$483k GPT-4o-mini

LFM2.5-VL-450M, self-hosted · per year

\$660 1× box, one-time hw

Processing **126M image-inferences/year** (36M new + 90M from 3× re-processing). **GPT-4o-mini** bills **\$483k/yr, every year**. On-device is a **\$660/yr** hardware line that re-processing doesn't move — about **731× less**, and the images never leave your network.

Assumptions · API = input tokens only at 2026 list price (output extra). On-device = 130M img/device/yr (242 ms/img, Jetson Orin, Q4_0; batched GPU is higher), \$1500/device over 3 yrs + \$160/yr power. Marginal cost per extra image or re-run: \$0.

The point is not that the API is unaffordable — at the cheapest tier it often isn't. The point is that on-device removes the meter entirely, so **re-processing, enrichment, and full-corpus coverage become free decisions** instead of budget fights — and the data never leaves. Cost is only one of four axes; the other three (data residency, throughput ceilings, ownership) the calculator can't price.

WHY A 450M MODEL IS THE RIGHT TOOL

Good enough to triage. Built for detection, description, counting — not transcription.

LFM2.5-VL-450M (released April 2026) pairs a 350M LFM2.5 hybrid-conv backbone with an 86M SigLIP2 vision encoder, handles images at native resolution up to 512×512, and added **bounding-box grounding and function-calling** in this generation. Against the obvious same-class peer, SmolVLM2-500M, it wins on every pipeline-relevant benchmark:

Object grounding / bounding boxes (RefCOCO-M)

81.3

new in 2.5 · 0 → 81 vs predecessor

Counting objects (CountBench)

73.3

+25.7 vs LFM2-VL-450M

Real-world scene Q&A (RealWorldQA)

58.4

vs SmolVLM2-500M 49.9

Hallucination resistance (POPE)

86.9

vs SmolVLM2-500M 82.7

Also: MMStar 43.0 (vs 38.2), MMBench-dev-en 60.9 (vs 52.3), MMVet 41.1. It runs in **219 MB** at 4-bit and returns a 512×512 answer in **~242 ms on a single Jetson Orin** — batched GPU serving multiplies that throughput by an order of magnitude.

Where it is NOT the answer — so you don't get burned in week one

Dense text extraction. OCRBench 684 trails Qwen2.5-VL-7B's 864 (both 0–1000). If a pipeline stage's job is reading paragraphs off documents or invoices, route those frames to an OCR specialist or a larger model. This model earns its seat on **the triage layer — detect, describe, tag, count, route — not the transcription layer**. Pitching it as an OCR engine is how you lose the eval.

THE SPINE · OWN A SPECIALIST, DON'T RENT A GENERALIST

Small enough to fine-tune in an afternoon, then it's yours — frozen, private, everywhere.

At 450M, you can fully fine-tune on a single GPU in hours — Liquid's own model card explicitly recommends fine-tuning for narrow use cases. So you don't bend your pipeline around a generalist's quirks; you bend a **specialist to your taxonomy**, pin the weights, and run the exact same artifact across your whole batch fleet.

No silent version bumps under you. No rate-limit tickets. No per-call meter. No model you're forbidden to ship into an air-gapped or on-prem environment. The model decision is one you control — which, not coincidentally, is the slot Liquid is built to win.

- **Yours:** weights you hold, version you freeze — reproducible labels across re-runs
- **Private:** point it at PHI/PII/regulated imagery; nothing leaves the boundary
- **Portable:** Transformers, vLLM, SGLang, llama.cpp, MLX, ONNX, LEAP — CPU to Jetson to phone
- **Proven on-device:** shipping embedded in Mercedes-Benz MB.OS in-car intelligence (2026)

Procurement note (be the one who flags it): ships under the **LFM Open License v1.0**, not Apache/MIT. Have legal read the commercial terms before fleet deployment.

THE FIELD, HONESTLY

What the buyer is actually choosing between

Scored for the *corpus-triage* job specifically — not a general leaderboard. Benchmarks are vision-side; OCRBench is 0–1000.

OPTION	SIZE	WHERE IT RUNS	\$ / 1K IMG*	MMSTAR	G
LFM2.5-VL-450M	0.45B	CPU · edge · phone · GPU	≈ \$0 marginal	43.0	R

OPTION		SIZE	WHERE IT RUNS	\$ / 1K IMG*	MMSTAR	G
Qwen2.5-VL-3B	self-host	~3B	GPU (heavier/img)	≈ \$0 marginal	55.9	s
SmolVLM2-500M	self-host	0.5B	CPU · edge · GPU	≈ \$0 marginal	38.2	li
Gemini 2.5 Flash-Lite	API	—	Cloud only	\$0.05	—	c
GPT-4o-mini	API	—	Cloud only	\$3.83	—	c
Silicon-partner VLM	on-device	~3B	One NPU SKU	≈ \$0 marginal	—	v

*Input-token list price, 2026. **The honest read:** Qwen2.5-VL-3B is the real self-host rival and scores higher on general VQA — but it's ~6× the parameters (heavier batch throughput, fatter GPUs) and its 3B tier is Research-licensed. SmolVLM2-500M matches on size and license openness but lacks grounding and trails on scene understanding. The APIs win on raw capability and lose on the two things a corpus pipeline cares about most: **marginal cost at re-processing scale, and data residency.** LFM2.5-VL-450M is the point on this map where *small, capable-enough, grounded, ownable, and private* all hold at once.

INTEGRATION RISK · REMOVED

Pointing it at images is five lines.

```
# CPU or GPU, fully offline – same path that runs in your batch workers
brew install llama.cpp
hf download LiquidAI/LFM2.5-VL-450M-GGUF
llama-mtmd-cli -m LFM2.5-VL-450M-Q4_0.gguf --mmproj mmproj-LFM2.5-VL-450M-Q8_0.gguf \
  --image frame.jpg -p "List objects, their boxes, and a one-line caption as JSON."
# drop -ngl for pure CPU · swap in your fine-tuned weights · wrap in vLLM/SGLang for b
```

Day-zero support in Transformers ≥5.2, vLLM 0.19, SGLang, llama.cpp, MLX, ONNX Runtime, and the LEAP Edge SDK (iOS/Android/desktop/Linux/Windows). GGUF + mmproj is the recommended path; ExecuTorch coexists.

A test designed to kill the idea fast if it's wrong.

- 1 Take **100k images** from one real corpus you already struggle to keep labeled.
- 2 Fine-tune LFM2.5-VL-450M on your taxonomy on a **single GPU** (hours, not weeks).
- 3 Run it **locally** over the set; capture tags, boxes, captions, triage routing.
- 4 Measure **triage precision/recall** and **cost-per-million** against your current API or human-labeling baseline.

If it misses your bar, you've spent a GPU-week and learned it. If it clears it, you've removed a **re-processing line item that compounds every quarter** and a **data-governance blocker** — in the same move.

WHAT THIS IS

An independent technical field note on LFM2.5-VL-450M for data & ML platform buyers, built around the one job a sub-half-billion-parameter VLM is uniquely suited to: corpus-scale visual triage. Not an official Liquid AI asset.

Honesty stance: every quantitative claim is sourced below; the model's OCR weakness, its non-Apache license, and the limits of the cost argument are stated in the body rather than buried.

PRIMARY SOURCES

1. LFM2.5-VL-450M model card & benchmarks — huggingface.co/LiquidAI/LFM2.5-VL-450M
2. GGUF weights / quant sizes — [.../LFM2.5-VL-450M-GGUF](https://huggingface.co/LiquidAI/LFM2.5-VL-450M-GGUF)
3. Launch + architecture — liquid.ai/blog/lfm2-5-vl-450m
4. On-device latency & deploy docs — docs.liquid.ai
5. Mercedes-Benz design win (Apr 2026) — Business Wire
6. API list prices & image tokenization — OpenAI / Google / Anthropic pricing pages, 2026
7. Competitor cards — Qwen2.5-VL, SmolVLM2 (HF)